



HYPERXITE 11

[Controls]

Graphical User Interface

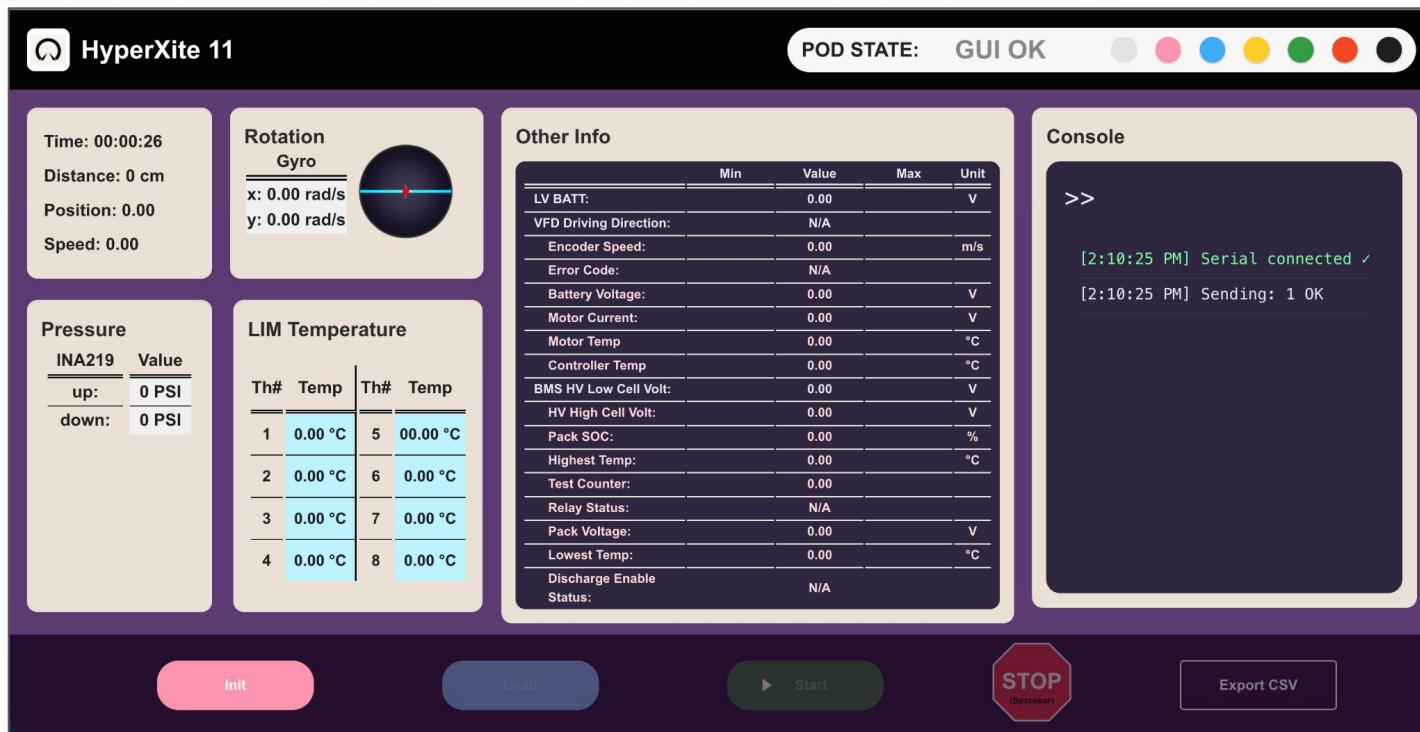


Figure 1. GUI Design

Key Analyses



- Choice of microcontroller: **STM32 Nucleo-144** vs. Raspberry Pi 5
 - Pi is great for raw computing power and multitasking (prev. HX design choice)
 - STM32 consume less power and we can design our own operating system
- STM32 offers native CAN and ADC support
 - Less external components/sensors for acquiring data

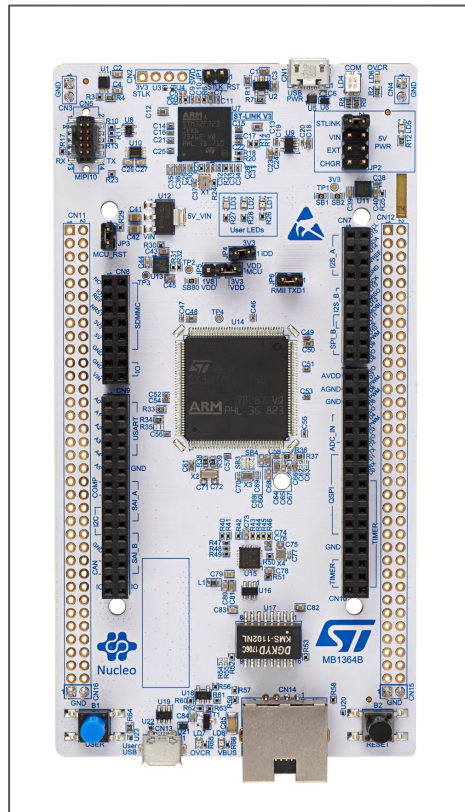


Figure 2. STM32 Nucleo-H753ZI

Constraints and Risks



- System constraints
 - **Limited onboard networking:** STM32 lacks native high-level networking support, constraining direct communication with external systems
 - **Real-time telemetry demands:** The system should support high-speed, low-latency data communication for live GUI visualization
- Risks:
 - **Sensor reliability and detection logic:** Inaccurate sensor readings or missed threshold values by the FSM could prevent timely fault detection, leading to unsafe system behavior

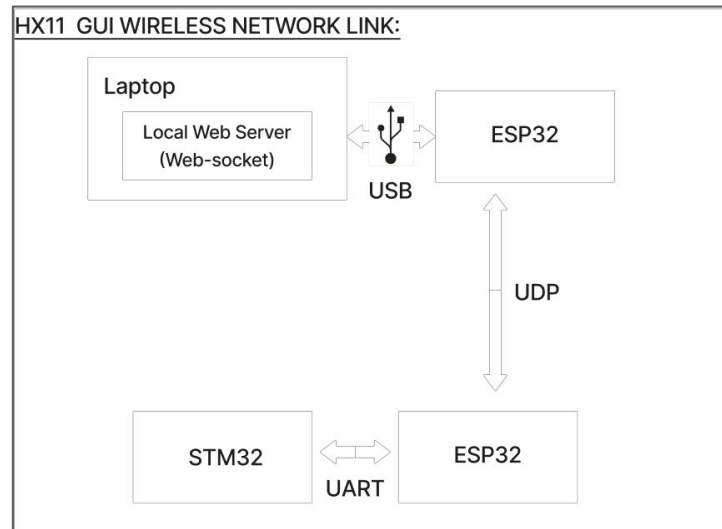


Figure 3. GUI Wireless Network Link

Finite State Machine Design

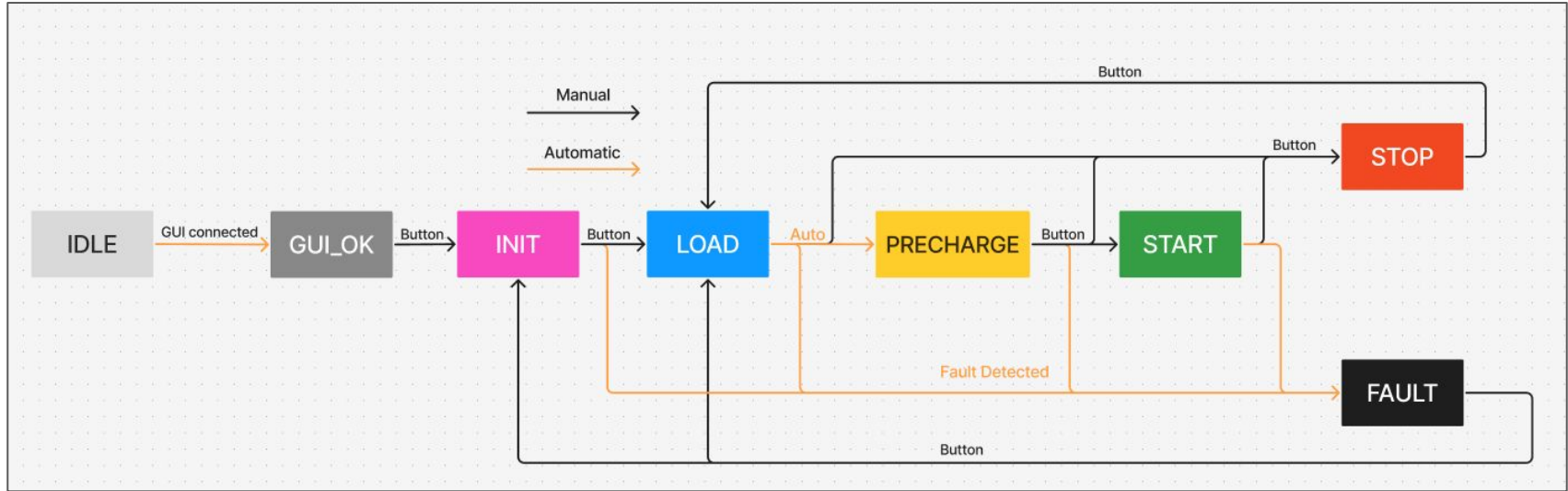


Figure 4: Finite State Machine

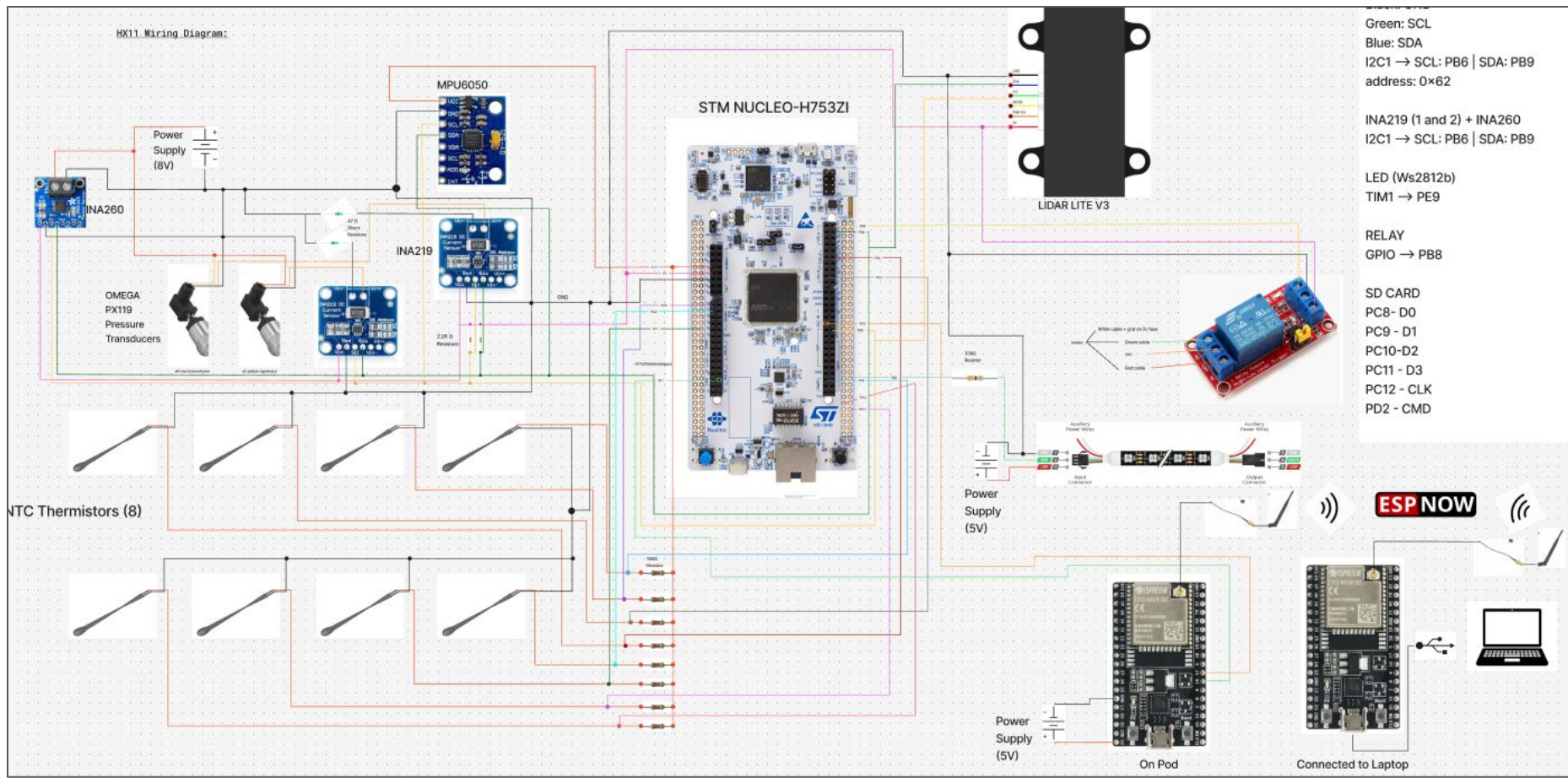


Figure 5. Wiring Diagram

Final Design



Figure 6: Control Board PCB 3D Model

